Roadway Infrastrastructure Economic Resilience

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Problem Statement

Current pavement selection decisions fail to evaluate pavement alternatives as a long-term system with external interactions. The investment decision has financial impacts that compound over decades through maintenance cycles, market exposure, and infrastructure disruptions. This project aims to improve the economic resilience of the City of Charlotte's roadway infrastructure.

Objectives

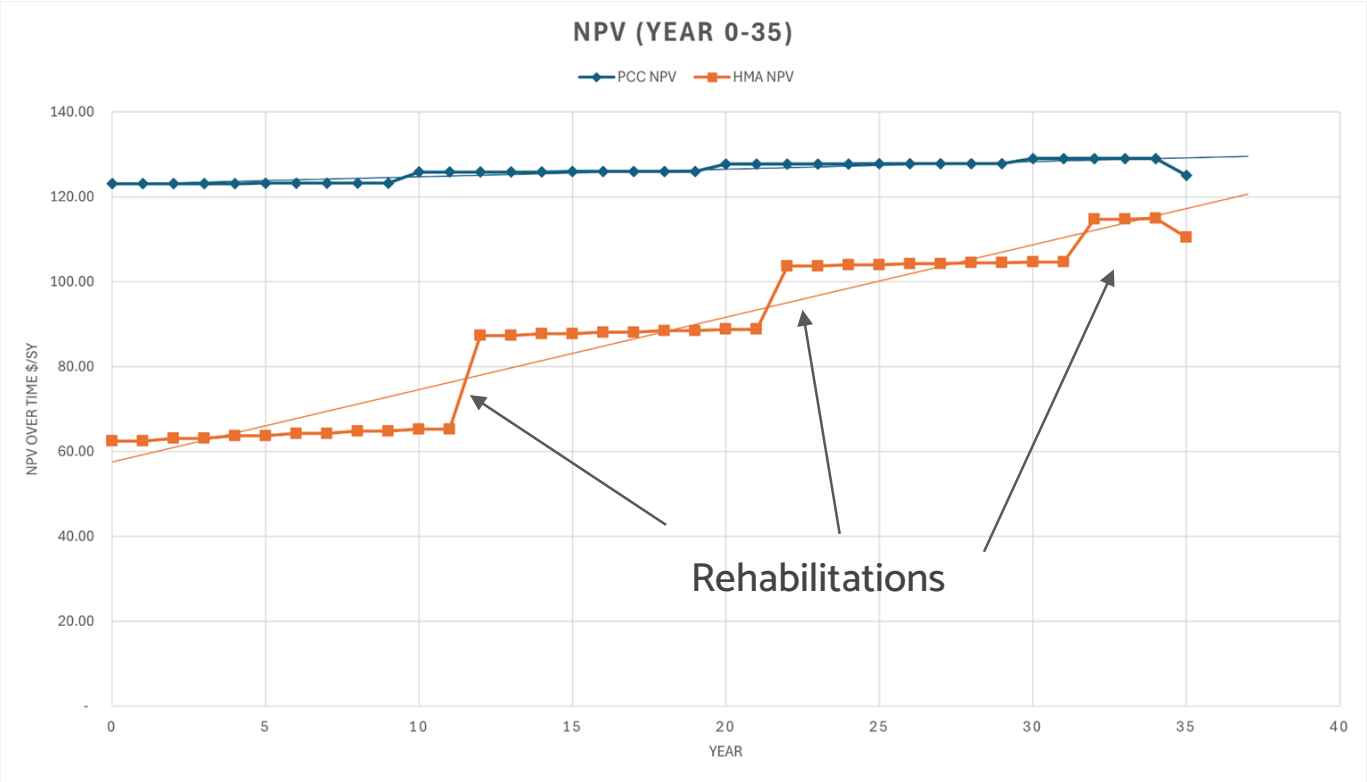
1. Quantify the pavement life cycle costs including initial costs, maintenance, major repairs and residual value
2. Apply capital costing methods, including NPV, to support comparison of pavement alternatives under different economic and operational scenarios
3. Evaluate uncertainty using scenario-based analysis
4. Quantify societal and sustainability impacts
5. Provide recommendations for Charlotte infrastructure decisions on future pavement selection and financial planning.

NPV Estimation

- MARR: 4.1%
  - 30 year nominal interest rate on treasury notes (Office of Management and Budget, 2026)
- Study Life: 35 years
- PCC Useful Life ≅ 40 Years
- HMA Useful Life ≅ 35 Years
- Residual value = (Years since last major investment/Uselful life) * Major investment Price

Alternate	Initial Cost$/SY	O&M #1	$/SY	Interval	O&M #2	$/SY	Interval
PCC	123.13	Crack Repair	0.70	2 years	1.5" mill and repave	35.36	10 years
HMA	62.46	Joint Sealing	0.20	5 years	Diamond Grinding	3.86	12 year initial then 10 years

NPV Continued



35 Year LCA		
Year	PCC	HMA
0	(123.08)	(62.46)
1	-	-
2	-	($0.70)
3	-	-
4	-	($0.70)
5	($0.20)	-
6	-	($0.70)
7	-	-
8	-	($0.70)
9	-	-
10	($3.86)	($0.70)
11	-	-
12	-	($35.36)
13	-	-
14	-	($0.70)
15	($0.20)	-
16	-	($0.70)
17	-	-
18	-	($0.70)
19	-	-
20	($3.86)	($0.70)
21	-	-
22	-	($35.36)
23	-	-
24	-	($0.70)
25	($0.20)	-
26	-	($0.70)
27	-	-
28	-	($0.70)
29	-	-
30	($3.86)	($0.70)
31	-	-
32	-	($35.36)
33	-	-
34	-	($0.70)
35	15.58	17.68

NPV	$ (125.04)	$ (110.46)

Sensitivity Analysis

Base Case

6 Parameters Tested

P1	PCC Initial Cost	$123.08/SY
P2	HMA Initial Cost	$62.46/SY
P3	PCC Maintenance PV	$5.91/SY
P4	HMA Maintenance PV	$52.47/SY
P5	MARR	4.0%
P6	Salvage Value	PCC $15.58 / HMA $17.68

PCC Present Value Cost

$125.04

per square yard ($/SY) | 35 years @ 4%

HMA Present Value Cost

$110.46

per square yard ($/SY) | 35 years @ 4%

HMA Savings over PCC

$14.59

per square yard — HMA is preferred alternative

HMA is the preferred pavement type — it costs $14.59/SY less in present value over 35 years

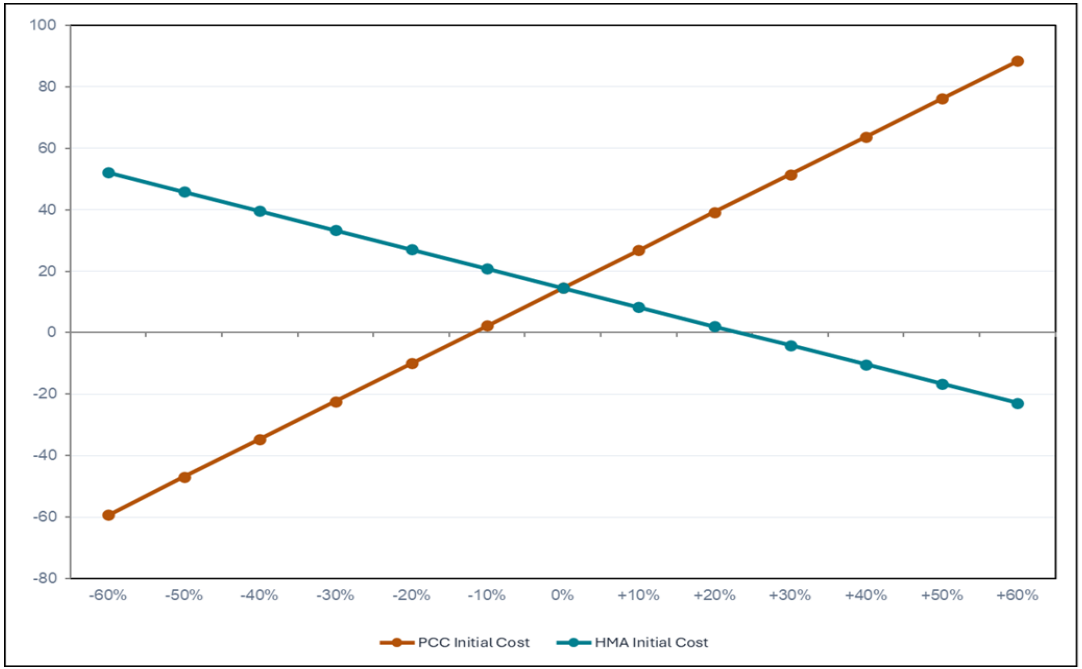
Sensitivity Analysis

One at A Time - Sensitivity Analysis

One-at-a-Time Sensitivity Analysis			
Rank	**Parameter**	**Abs. slope**	**Decision Reversal**
1	PCC Initial Cost	123.08	Yes @ -11.85%
2	HMA Initial Cost	62.46	Yes @ 23.35%
3	HMA Maintenance PV	52.47	Yes @ 27.80%
4	MARR	35.32	Yes @ -36.65%
5	PCC Maintenance PV	5.91	No in ±60%
6	Salvage Value	0.53	No in ±60%

Sensitivity Analysis

One at A Time - Sensitivity Analysis



Key Findings

PCC Initial Cost
Reversal at –11.85%
If PCC cost drops below $108.49/SY, PCC becomes cheaper. High sensitivity.

HMA Initial Cost
Reversal at +23.35%
HMA stays preferred unless its cost rises above $77.05/SY.

HMA is preferred in this case, with savings of $14.59/SY over PCC,. The decision is most sensitive to PCC initial cost and HMA initial cost

Sensitivity Analysis

Two at A Time - Sensitivity Analysis

Rows = HMA Maint. % change | Columns = HMA Initial Cost % change													
HMA Maint. \ HMA Init.	-60%	-50%	-40%	-30%	-20%	-10%	0%	10%	20%	30%	40%	50%	60%
-60%	83.55	77.30	71.05	64.81	58.56	52.32	46.07	39.82	33.58	27.33	21.08	14.84	8.59
-50%	78.30	72.05	65.81	59.56	53.31	47.07	40.82	34.58	28.33	22.08	15.84	9.59	3.34
-40%	73.05	66.81	60.56	54.31	48.07	41.82	35.57	29.33	23.08	16.84	10.59	4.34	-1.90
-30%	67.81	61.56	55.31	49.07	42.82	36.57	30.33	24.08	17.83	11.59	5.34	-0.90	-7.15
-20%	62.56	56.31	50.07	43.82	37.57	31.33	25.08	18.83	12.59	6.34	0.09	-6.15	-12.40
-10%	57.31	51.07	44.82	38.57	32.33	26.08	19.83	13.59	7.34	1.09	-5.15	-11.40	-17.65
0%	52.06	45.82	39.57	33.33	27.08	20.83	14.59	8.34	2.09	-4.15	-10.40	-16.65	-22.89
10%	46.82	40.57	34.32	28.08	21.83	15.59	9.34	3.09	-3.15	-9.40	-15.65	-21.89	-28.14
20%	41.57	35.32	29.08	22.83	16.58	10.34	4.09	-2.15	-8.40	-14.65	-20.89	-27.14	-33.39
30%	36.32	30.08	23.83	17.58	11.34	5.09	-1.16	-7.40	-13.65	-19.89	-26.14	-32.39	-38.63
40%	31.08	24.83	18.58	12.34	6.09	-0.16	-6.40	-12.65	-18.90	-25.14	-31.39	-37.63	-43.88
50%	25.83	19.58	13.34	7.09	0.84	-5.40	-11.65	-17.90	-24.14	-30.39	-36.64	-42.88	-49.13
60%	20.58	14.34	8.09	1.84	-4.40	-10.65	-16.90	-23.14	-29.39	-35.64	-41.88	-48.13	-54.38
Positive savings = HMA is cheaper than PCC. Negative savings = PCC is cheaper.													

This shows that HMA is cheaper with savings of $14.59/SY. But, when HMA initial and maintenance costs increase together, the savings decrease and the decision can reverse in favour of PCC.

Scenario Analysis: Usage-Based Rehabilitation

Traffic loading determines how often each pavement type needs rehabilitation. More rehab events = higher total cost for HMA.

Rehabilitation Intervals

Scenario	Prob.	HMA (yrs)	PCC (yrs)	HMA Events
High Use	25%	8	18	**4**
Average Use	50%	11	24	**3**
Low Use	25%	14	30	**2**

PCC diamond grinding intervals are 2-3x longer than HMA overlay intervals across all scenarios.

35-Year NPV Results ($/SY)

Scenario	HMA ($/SY)	PCC ($/SY)	Difference	Lower Cost
High Use	$137.49	$125.62	**($11.87)**	**PCC**
Average Use	$116.48	$125.22	**$8.74**	**HMA**
Low Use	$101.11	$124.91	**$23.80**	**HMA**
Prob.-Weighted Expected	**$117.89**	**$125.24**	**$7.35**	**HMA**

HMA lower expected cost by $7.35/SY, but PCC wins if high-use conditions develop.

(Robbins, 2018)

Scenario Analysis: Oil Market Risk + Combined Recommendation

Asphalt binder is petroleum-derived. Every future HMA resurfacing event carries oil price risk that PCC does not share.

Oil Market NPV Results — Average Use Intervals ($/SY)

Oil Condition	HMA Mult.	HMA NPV ($/SY)	PCC NPV ($/SY)	Lower Cost
High Cost (+100%)	**1.70x**	$193.52	$125.22	**PCC**
Average (Base)	1.00x	$116.48	$125.22	**HMA**
Low Cost (-50%)	**0.50x**	$61.45	$125.22	**HMA**
Prob.-Weighted Expected	**1.05x**	**$121.98**	**$125.22**	**HMA**

HMA NPV swings $132/SY depending on oil market conditions. PCC NPV is flat across all conditions.

RECOMMENDATION

HMA (Asphalt)

Expected lower-cost alternative across both scenario frameworks

Usage Scenarios

HMA $117.89 vs PCC $125.24

HMA saves $7.35/SY

Oil Market Scenarios

HMA $121.98 vs PCC $125.22

HMA saves $3.24/SY

Societal Benefits

Maintenance and rehabilitation activities create work zones, reducing roadway capacity and increasing travel time for users.

Based on Federal Highway Administration Life-Cycle Cost Analysis (LCCA) guidance, societal impacts can be quantified using user delay costs (U.S. Department of Transportation, 2025), calculated as:

Total Delay Hours x Value of Time ($/hour).

These societal impacts are not included in the base cost analysis but provide an important measure of public benefit and can influence overall infrastructure investment decisions.

Societal Benefits

Average Daily Traffic (ADT)	20,000 vehicles/day (NCDOT, n.d.)
Delay per vehicle*	15 minutes (0.25 hours)
Duration of work*	Asphalt - 5 days Concrete - 2 days
Value of time	$21.10/hour (U.S. Department of Transportation, 2025)

Year	Asphalt User Cost	Concrete User Cost
12	$527,500	0
15	0	$211,000
24	$527,500	0

Asphalt User Cost per Event
=20,000×0.25×5×21.10=$527,500

Concrete User Cost per Event
=20,000×0.25×2×21.10=$211,000

*Delay, duration, and maintenance intervals are preliminary assumptions and will be refined using literature and project-specific data in the final analysis.

**These societal impacts are not included in the base cost analysis but will be incorporated into the final NPV model to evaluate total economic impact.

Sustainability

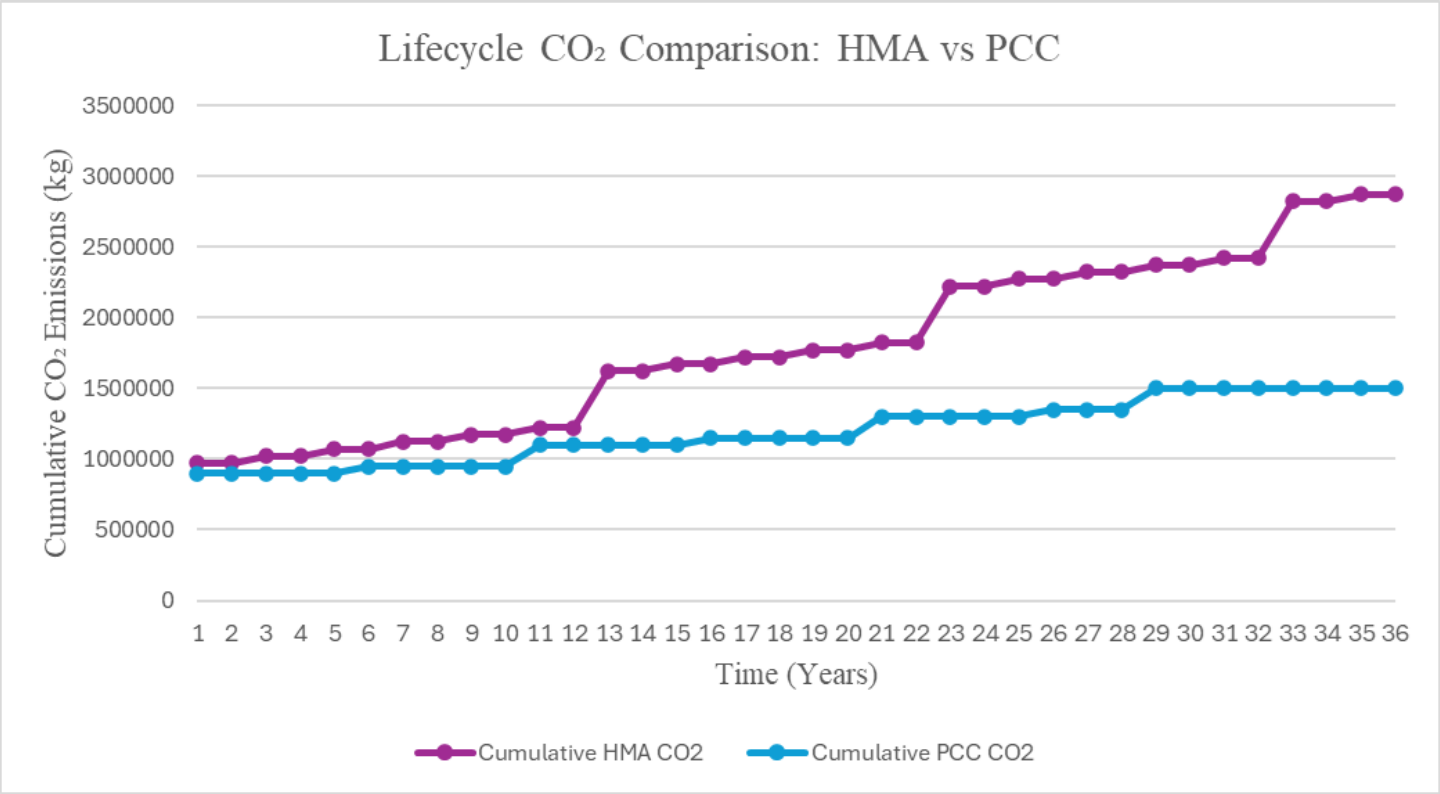
Factors :
Initial construction emissions
Maintenance & rehabilitation activities
Frequency of maintenance cycles
Cumulative lifecycle CO_2 emissions

HMA total emissions ≈ 2.87M CO_2 over 35 years
PCC total emissions ≈ 1.50M CO_2 over 35 years
Higher emissions in HMA are driven by frequent resurfacing cycles

Cost v/s CO2 Tradeoff:

	CO2	Cost
HMA	2872000	$110.46
PCC	1500000	$125.04

Sustainability



Next Steps

- Further refine NPV analysis and interpret data
- Create a cost benefit analysis with pavement alternative findings
- Societal impact inputs to be reviewed within the cash flow model
- Compile findings and determine economic implications for the city of Charlotte funding allocations
- Justify recommendations for applications of pavement alternatives
- Expand the analysis with additional uncertainty methods, such as optimistic–pessimistic analysis
- Explore potential applications of AI, such as predictive maintenance, to enhance pavement lifecycle cost analysis

Sources

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